

Vernier Calipers & Screw Gauge — Advanced Problem Set

Nine Education IIT Academy | Class 11 Physics | 50 Problems

Least Count (Vernier): $LC = 1 \text{ MSD} / n$ where n = number of vernier scale divisions

Least Count (Screw Gauge): $LC = \text{Pitch} / (\text{no. of circular divisions})$ where Pitch = distance advanced per full rotation

Section A — Non-Standard Vernier Calipers

1. Vernier: 20 VSD, 1 MSD = 1 mm. Jaws closed: 12th division coincides, right of main scale zero. Measuring a block: MSR = 4.1 cm, 7th division coincides. Find LC, zero error, and corrected length.
2. Vernier: 25 VSD, 1 MSD = 1 mm. Jaws closed: 18th division coincides, right of main scale zero. Measuring a rod: MSR = 3.5 cm, 10th division coincides. Find LC, zero error, and corrected length.
3. Vernier: 50 VSD, 1 MSD = 1 mm. Jaws closed: 30th division coincides, left of main scale zero. Measuring a cylinder: MSR = 7.0 cm, 22nd division coincides. Find LC, zero error, and corrected diameter.
4. Vernier: 20 VSD, 1 MSD = 1 mm. Zero error = +0.010 cm. Measuring a metal block: MSR = 1.6 cm, 15th division coincides. Find LC and corrected length.
5. Vernier: 25 VSD, 1 MSD = 1 mm. Jaws closed: 5th division coincides, left of main scale zero. Measuring a marble: MSR = 2.5 cm, 20th division coincides. Find LC, zero error, and corrected diameter.
6. Vernier: 50 VSD, 1 MSD = 1 mm. Jaws closed: 45th division coincides, left of main scale zero. Measuring a plate: MSR = 5.0 cm, 8th division coincides. Find LC, zero error, and corrected thickness.
7. Vernier: 20 VSD, 1 MSD = 1 mm. Jaws closed: 16th division coincides, right of main scale zero. Measuring a sphere: MSR = 4.8 cm, 3rd division coincides. Find LC, zero error, and corrected diameter.
8. Vernier: 25 VSD, 1 MSD = 1 mm. Zero error = -0.016 cm. Measuring a rod: MSR = 6.0 cm, 10th division coincides. Find LC and corrected length.

9. Vernier: 50 VSD, 1 MSD = 1 mm. Jaws closed: 4th division coincides, right of main scale zero. Measuring a cube's side: MSR = 3.0 cm, 40th division coincides. Find LC, zero error, and corrected side length.
10. Vernier: 20 VSD, 1 MSD = 1 mm. Jaws closed: 2nd division coincides, left of main scale zero. Measuring a rod: MSR = 5.2 cm, 18th division coincides. Find LC, zero error, and corrected length.
11. Vernier: 25 VSD, 1 MSD = 1 mm. Jaws closed: 22nd division coincides, right of main scale zero. Measuring an object: 0th vernier division coincides, MSR = 3.8 cm. Find LC, zero error, and corrected length.
12. Vernier: 50 VSD, 1 MSD = 1 mm. Zero error = -0.006 cm. Measuring a coin: MSR = 2.2 cm, 35th division coincides. Find LC and corrected diameter.

Section B — Multi-Step Screw Gauge

13. Screw gauge advances 3 mm in 6 rotations; 50 circular divisions. Jaws closed: 48th division on the reference line (thimble zero above). Measuring a wire: MSR = 1 mm, 26th division on the reference line. Find pitch, LC, zero error, and corrected diameter.
14. Advances 4 mm in 8 rotations; 100 circular divisions. Jaws closed: 3rd division on the reference line (thimble zero below). Measuring a ball bearing: MSR = 5 mm, 72nd division. Find pitch, LC, zero error, and corrected diameter.
15. Pitch = 1 mm; 200 circular divisions. Jaws closed: 197th division on the reference line (thimble zero above). Measuring a wire: MSR = 0 mm, 85th division. Find LC, zero error, and corrected diameter.
16. Pitch = 0.5 mm; 50 circular divisions. Jaws closed: 3rd division on the reference line (thimble zero below). Measuring a ball: 4.0 mm mark visible, 4.5 mm half-mark also visible, 22nd division. Find LC, zero error, and corrected diameter.
17. Pitch = 0.5 mm; 50 circular divisions. Jaws closed: 47th division on the reference line (thimble zero above). Measuring a wire: 2.0 mm mark visible, 2.5 mm half-mark NOT visible, 35th division. Find LC, zero error, and corrected diameter.
18. Advances 2 mm in 4 rotations; 100 circular divisions. Jaws closed: 5th division on the reference line (thimble zero below). Measuring a sphere: MSR = 3 mm, 64th division. Find pitch, LC, zero error, and corrected diameter.

- 19.** Pitch = 1 mm; 200 circular divisions. Jaws closed: 195th division on the reference line (thimble zero above). Measuring a plate: MSR = 2 mm, 120th division. Find LC, zero error, and corrected thickness.
- 20.** Pitch = 0.5 mm; 50 circular divisions. Jaws closed: 4th division on the reference line (thimble zero below). Measuring a wire: 0.0 mm mark visible, 0.5 mm half-mark also visible, 18th division. Find LC, zero error, and corrected diameter.
- 21.** Advances 5 mm in 10 rotations; 50 circular divisions. Jaws closed: 49th division on the reference line (thimble zero above). Measuring a ball: MSR = 6 mm, 15th division. Find pitch, LC, zero error, and corrected diameter.
- 22.** Pitch = 1 mm; 100 circular divisions. Jaws closed: 96th division on the reference line (thimble zero above). Measuring a thick wire: MSR = 4 mm, 10th division. Find LC, zero error, and corrected diameter.
- 23.** Pitch = 0.5 mm; 50 circular divisions. Jaws closed: 2nd division on the reference line (thimble zero below). Measuring a cylinder: 3.0 mm mark visible, 3.5 mm half-mark also visible, 40th division. Find LC, zero error, and corrected diameter.
- 24.** Advances 6 mm in 12 rotations; 200 circular divisions. Jaws closed: 198th division on the reference line (thimble zero above). Measuring a wire: MSR = 1 mm, 160th division. Find pitch, LC, zero error, and corrected diameter.
- 25.** Pitch = 1 mm; 100 circular divisions. Jaws closed: 6th division on the reference line (thimble zero below). Measuring a wire: MSR = 0 mm, 54th division. Find zero error, corrected diameter, and the percentage error in its cross-sectional area. (Take LC as the absolute error in diameter.)
- 26.** Two screw gauges measure the same wire. Find LC of each, corrected diameter from each, and state which is more precise:
- Gauge A: pitch = 1 mm, 100 div, ZE = +0.02 mm. MSR = 0 mm, 52nd division.
 - Gauge B: pitch = 0.5 mm, 100 div, ZE = -0.01 mm. MSR = 0 mm, 98th division.
- 27.** Pitch = 0.5 mm; 50 circular divisions. Jaws closed: 3rd division on the reference line (thimble zero below). Measuring a ball: 7.0 mm mark visible, 7.5 mm half-mark also visible, 28th division. Find LC, zero error, and corrected diameter.

- 28.** Pitch = 0.5 mm; 50 circular divisions. Jaws closed: 48th division on the reference line (thimble zero above). Measuring a rod: 3.0 mm mark visible, 3.5 mm half-mark NOT visible, 20th division. Find LC, zero error, and corrected diameter.

Section C — Comparison & Repeated Readings

- 29.** Two vernier calipers measure the same rod. Find LC of each, corrected reading from each, and state which is more precise:
- Vernier A (10 VSD, 1 MSD = 1 mm): ZE = +0.02 cm. MSR = 2.3 cm, 8th division coincides.
 - Vernier B (20 VSD, 1 MSD = 1 mm): ZE = -0.01 cm. MSR = 2.3 cm, 10th division coincides.
- 30.** Screw gauge: pitch = 1 mm, 100 div, ZE = -0.03 mm. Five readings of a wire (all MSR = 0 mm), circular divisions: 46, 48, 47, 49, 45. Find LC, corrected readings, and express mean diameter as (mean $\pm$ mean absolute error) mm.
- 31.** Screw gauge: pitch = 0.5 mm, 50 div, ZE = -0.04 mm. Five readings of a ball (all MSR = 5 mm, half-mark visible), circular divisions: 28, 30, 29, 31, 27. Find LC, corrected mean diameter, and mean absolute error.
- 32.** Vernier (20 VSD, 1 MSD = 1 mm). ZE = +0.015 cm. Observed diameter of a sphere = 3.115 cm. Find the corrected diameter and the percentage error in volume. (Take LC as absolute error in diameter.)
- 33.** Screw gauge: pitch = 1 mm, 100 div, ZE = +0.04 mm. Observed diameter of a wire = 1.54 mm; length $L = (50.0 \pm 0.1)$ mm. Find corrected diameter and the percentage error in the volume of the wire.
- 34.** Screw gauge: pitch = 1 mm, 200 div. Jaws closed: 3rd division on the reference line (thimble zero below). Measuring a rod: MSR = 2 mm, 140th division. Mass = (0.500 ± 0.001) g; length = (50.0 ± 0.1) mm. Find LC, ZE, corrected diameter, and percentage error in density. [$\rho = 4M/(\pi d^2 L)$]
- 35.** A sphere is measured by two instruments. Find the corrected diameter from each:
- Vernier (50 VSD, 1 MSD = 1 mm): ZE = +0.004 cm. MSR = 0.4 cm, 30th division coincides.
 - Screw gauge (pitch = 0.5 mm, 50 div): ZE = -0.02 mm. 4.0 mm mark, 4.5 mm half-mark visible, 10th division.

- 36.** Vernier (25 VSD, 1 MSD = 1 mm). $ZE = -0.020$ cm. Three readings of a rod: (MSR = 4.5 cm, 10th div), (MSR = 4.5 cm, 12th div), (MSR = 4.5 cm, 11th div). Find LC, corrected readings, mean length, and mean absolute error.
- 37.** Screw gauge: pitch = 0.5 mm, 50 div. Jaws closed: 46th division on the reference line (thimble zero above). Ball measured in 4 positions (all MSR = 5 mm, half-mark visible), circular divisions: 25, 27, 24, 26. Find LC, ZE, corrected mean diameter, and mean absolute error.
- 38.** A brass cube's side is measured by two instruments. Find LC of each, corrected side length from each, and state which is more precise:
- Vernier (50 VSD, 1 MSD = 1 mm): $ZE = -0.004$ cm. MSR = 2.0 cm, 42nd division coincides.
 - Screw gauge (pitch = 1 mm, 100 div): $ZE = +0.02$ mm. MSR = 20 mm, 84th division.
- 39.** Screw gauge: pitch = 1 mm, 200 div. Jaws closed: 196th division on the reference line (thimble zero above). Cylinder diameter at 3 positions (all MSR = 3 mm), circular divisions: 140, 136, 138. Find LC, ZE, corrected readings, mean diameter, and mean absolute error.

Section D — Applied & Tricky Problems

- 40.** Vernier (20 VSD, 1 MSD = 1 mm). $ZE = +0.010$ cm. Corrected outer diameter of a pipe = 3.065 cm; corrected inner diameter = 2.460 cm. Find the wall thickness and its absolute error. (Take LC as the absolute error in each measurement.)
- 41.** Screw gauge: pitch = 1 mm, 100 div, $ZE = +0.03$ mm. Observed diameter of a rod = 3.52 mm. Length measured by a vernier (10 VSD, 1 MSD = 1 mm, $ZE = +0.02$ cm): MSR = 6.2 cm, 5th division coincides. Find corrected diameter, corrected length, and volume of the rod. ($\pi = 3.14$)
- 42.** Screw gauge: pitch = 0.5 mm, 50 div, $ZE = -0.01$ mm. Observed diameter of a wire = 0.48 mm; mass = (1.850 ± 0.001) g; length = (200.0 ± 0.5) mm. Find corrected diameter and the percentage error in density.

- 43.** Screw gauge: pitch = 1 mm, 100 div, ZE = +0.05 mm. Diameter of a wire measured in two perpendicular orientations — Horizontal: MSR = 0 mm, 50th division. Vertical: MSR = 0 mm, 48th division. Find both corrected diameters, their average, and state what the difference suggests about the wire.
- 44.** Vernier (25 VSD, 1 MSD = 1 mm). ZE = -0.012 cm. Outer diameter: MSR = 2.0 cm, 20th division. Inner diameter: MSR = 1.5 cm, 15th division. Find LC, both corrected diameters, and the cross-sectional area of the pipe material. ($\pi = 3.14$)
- 45.** Vernier (20 VSD, 1 MSD = 1 mm). Actual ZE = +0.020 cm (student forgot to check). Student records MSR = 4.0 cm, 12th division. (a) What does the student report? (b) What is the true corrected length? (c) By how much is the student's answer wrong?
- 46.** Screw gauge: pitch = 1 mm, 100 div. Actual ZE = -0.05 mm (student forgot). Student reads MSR = 3 mm, 50th division. (a) What does the student report? (b) What is the correct diameter? (c) What is the error in the student's answer?
- 47.** Screw gauge: pitch = 1 mm, 100 div. Jaws closed: 97th division on the reference line (thimble zero above). Two wires are measured — Wire 1: MSR = 1 mm, 42nd division. Wire 2: MSR = 0 mm, 98th division. Find LC, zero error, and corrected diameter of each wire.
- 48.** Screw gauge: pitch = 0.5 mm, 50 div. Jaws closed: 3rd division on the reference line (thimble zero below). Two balls measured — Ball A: 5.0 mm mark, 5.5 mm half-mark visible, 20th division. Ball B: 8.0 mm mark visible, 8.5 mm half-mark NOT visible, 38th division. Find LC, ZE, and corrected diameter of each ball.
- 49.** Vernier (20 VSD, 1 MSD = 1 mm). ZE = +0.020 cm. Observed length = 5.125 cm. (a) Find the corrected length. (b) A student ignores ZE and claims the true length lies within (5.120–5.130) cm. Is the student correct? Justify.
- 50.** A small ball is measured with both instruments. Find LC, ZE, and corrected diameter from each; then find the volume using the screw gauge reading. ($\pi = 3.14$)
- Vernier (20 VSD, 1 MSD = 1 mm): Jaws closed: 2nd division, right of zero. MSR = 1.1 cm, 12th division coincides.
 - Screw gauge (pitch = 0.5 mm, 50 div): Jaws closed: 49th division on reference line (above). 11.0 mm mark, 11.5 mm half-mark visible, 36th division.

Answer Key

Section A — Non-Standard Vernier Calipers

1. $LC = 0.005 \text{ cm}$; $ZE = +0.060 \text{ cm}$; $Corr = 4.075 \text{ cm}$
2. $LC = 0.004 \text{ cm}$; $ZE = +0.072 \text{ cm}$; $Corr = 3.468 \text{ cm}$
3. $LC = 0.002 \text{ cm}$; $ZE = -0.040 \text{ cm}$; $Corr = 7.084 \text{ cm}$
4. $LC = 0.005 \text{ cm}$; $Corr = 1.665 \text{ cm}$
5. $LC = 0.004 \text{ cm}$; $ZE = -0.080 \text{ cm}$; $Corr = 2.660 \text{ cm}$
6. $LC = 0.002 \text{ cm}$; $ZE = -0.010 \text{ cm}$; $Corr = 5.026 \text{ cm}$
7. $LC = 0.005 \text{ cm}$; $ZE = +0.080 \text{ cm}$; $Corr = 4.735 \text{ cm}$
8. $LC = 0.004 \text{ cm}$; $Corr = 6.056 \text{ cm}$
9. $LC = 0.002 \text{ cm}$; $ZE = +0.008 \text{ cm}$; $Corr = 3.072 \text{ cm}$
10. $LC = 0.005 \text{ cm}$; $ZE = -0.090 \text{ cm}$; $Corr = 5.380 \text{ cm}$
11. $LC = 0.004 \text{ cm}$; $ZE = +0.088 \text{ cm}$; $Corr = 3.712 \text{ cm}$
12. $LC = 0.002 \text{ cm}$; $Corr = 2.276 \text{ cm}$

Section B — Multi-Step Screw Gauge

13. $Pitch = 0.5 \text{ mm}$; $LC = 0.01 \text{ mm}$; $ZE = -0.02 \text{ mm}$; $Corr = 1.28 \text{ mm}$
14. $Pitch = 0.5 \text{ mm}$; $LC = 0.005 \text{ mm}$; $ZE = +0.015 \text{ mm}$; $Corr = 5.345 \text{ mm}$
15. $LC = 0.005 \text{ mm}$; $ZE = -0.015 \text{ mm}$; $Corr = 0.440 \text{ mm}$
16. $LC = 0.01 \text{ mm}$; $ZE = +0.03 \text{ mm}$; $Corr = 4.69 \text{ mm}$
17. $LC = 0.01 \text{ mm}$; $ZE = -0.03 \text{ mm}$; $Corr = 2.38 \text{ mm}$
18. $Pitch = 0.5 \text{ mm}$; $LC = 0.005 \text{ mm}$; $ZE = +0.025 \text{ mm}$; $Corr = 3.295 \text{ mm}$
19. $LC = 0.005 \text{ mm}$; $ZE = -0.025 \text{ mm}$; $Corr = 2.625 \text{ mm}$
20. $LC = 0.01 \text{ mm}$; $ZE = +0.04 \text{ mm}$; $Corr = 0.64 \text{ mm}$
21. $Pitch = 0.5 \text{ mm}$; $LC = 0.01 \text{ mm}$; $ZE = -0.01 \text{ mm}$; $Corr = 6.16 \text{ mm}$
22. $LC = 0.01 \text{ mm}$; $ZE = -0.04 \text{ mm}$; $Corr = 4.14 \text{ mm}$
23. $LC = 0.01 \text{ mm}$; $ZE = +0.02 \text{ mm}$; $Corr = 3.88 \text{ mm}$
24. $Pitch = 0.5 \text{ mm}$; $LC = 0.0025 \text{ mm}$; $ZE = -0.005 \text{ mm}$; $Corr = 1.405 \text{ mm}$
25. $ZE = +0.06 \text{ mm}$; $Corr d = 0.48 \text{ mm}$; $\% \text{ err in area} \approx 4.17\%$
26. A: $LC = 0.01 \text{ mm}$, $d = 0.50 \text{ mm}$ — B: $LC = 0.005 \text{ mm}$, $d = 0.500 \text{ mm}$ — B is more precise
27. $LC = 0.01 \text{ mm}$; $ZE = +0.03 \text{ mm}$; $Corr = 7.75 \text{ mm}$
28. $LC = 0.01 \text{ mm}$; $ZE = -0.02 \text{ mm}$; $Corr = 3.22 \text{ mm}$

Section C — Comparison & Repeated Readings

29. A: $LC = 0.01 \text{ cm}$, $d = 2.36 \text{ cm}$ — B: $LC = 0.005 \text{ cm}$, $d = 2.360 \text{ cm}$ — B more precise
30. $LC = 0.01 \text{ mm}$; $\text{Mean } d = (0.50 \pm 0.01) \text{ mm}$
31. $LC = 0.01 \text{ mm}$; $\text{Mean } d = (4.83 \pm 0.01) \text{ mm}$

32. Corr $d = 3.100$ cm; % err in $V \approx 0.48\%$

33. Corr $d = 1.50$ mm; % err in $V \approx 1.53\%$

34. LC = 0.005 mm; ZE = $+0.015$ mm; $d = 2.685$ mm; % err in $p \approx 0.77\%$

35. Vernier: 4.56 mm — Screw gauge: 4.62 mm

36. LC = 0.004 cm; Corr: $4.560, 4.568, 4.564$ cm; Mean = (4.564 ± 0.003) cm

37. LC = 0.01 mm; ZE = -0.04 mm; Mean $d \approx (5.80 \pm 0.01)$ mm

38. Vernier: LC = 0.02 mm, $d = 20.88$ mm — SG: LC = 0.01 mm, $d = 20.82$ mm — SG more precise

39. LC = 0.005 mm; ZE = -0.020 mm; Corr: $3.720, 3.700, 3.710$ mm; Mean = (3.710 ± 0.007) mm

Section D — Applied & Tricky Problems

40. Thickness = (0.303 ± 0.005) cm

41. Corr $d = 3.49$ mm; Corr $L = 62.30$ mm; $V \approx 596$ mm³

42. Corr $d = 0.49$ mm; % err in $p \approx 4.39\%$

43. $H = 0.45$ mm; $V = 0.43$ mm; Avg = 0.44 mm; cross-section is not perfectly circular

44. LC = 0.004 cm; $D(\text{out}) = 2.092$ cm; $D(\text{in}) = 1.572$ cm; Area ≈ 1.50 cm²

45. (a) 4.060 cm (b) 4.030 cm (c) Over-reads by 0.030 cm

46. (a) 3.50 mm (b) 3.55 mm (c) Under-reads by 0.05 mm

47. LC = 0.01 mm; ZE = -0.03 mm; Wire 1 = 1.45 mm; Wire 2 = 1.01 mm

48. LC = 0.01 mm; ZE = $+0.03$ mm; Ball A = 5.67 mm; Ball B = 8.35 mm

49. (a) 5.105 cm (b) No — 5.105 cm lies outside 5.120 – 5.130 cm; ignoring ZE gives a systematic error

50. Vernier $d = 11.50$ mm; SG $d = 11.87$ mm; $V \approx 875$ mm³